

PAPER_949: BCS Gap Equation in SCm Vacuum

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** bcs_superconductivity_uqff.py (BCSGapEquation) **Calculator:** BCSGapEquationCalc (CP4 #533) **CVW:** v2.0.0 compliant

Abstract

We derive the BCS energy gap equation in the SCm vacuum phonon framework. The gap Δ is determined self-consistently via $\Delta = (\hbar\omega_{\text{SCm}}/2) \cdot \tanh(\Delta/2k_B T) \cdot S_{26} \cdot (F_{U,\text{Bi}}/F_U)$, where $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz is the SCm phonon resonance frequency. This maps conventional BCS superconductivity to the UQFF vacuum structure, with the 26-layer buoyancy sum S_{26} replacing the Debye phonon spectrum.

1. Gap Equation

$$\Delta = \frac{\hbar\omega_{\text{SCm}}}{2} \tanh\left(\frac{\Delta}{2k_B T}\right) \cdot S_{26}([\text{SSq}]) \cdot \frac{F_{U,\text{Bi}}}{F_U}$$

Self-consistent solution via iterative fixed-point method converges in < 50 iterations at all temperatures.

2. Temperature Dependence

T (K)	Δ (eV)	Δ/Δ_0
0	Δ_0	1.000
1	$\approx \Delta_0$	≈ 1.000
100	reduced	< 1
T_c	0	0

3. Source Data

- **File:** bcs_superconductivity_uqff.py
 - **Session:** 214
 - **CP4 Class:** BCSGapEquationCalc (#533)
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References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Bardeen, J., Cooper, L.N. & Schrieffer, J.R. (1957) – Phys. Rev. 108, 1175